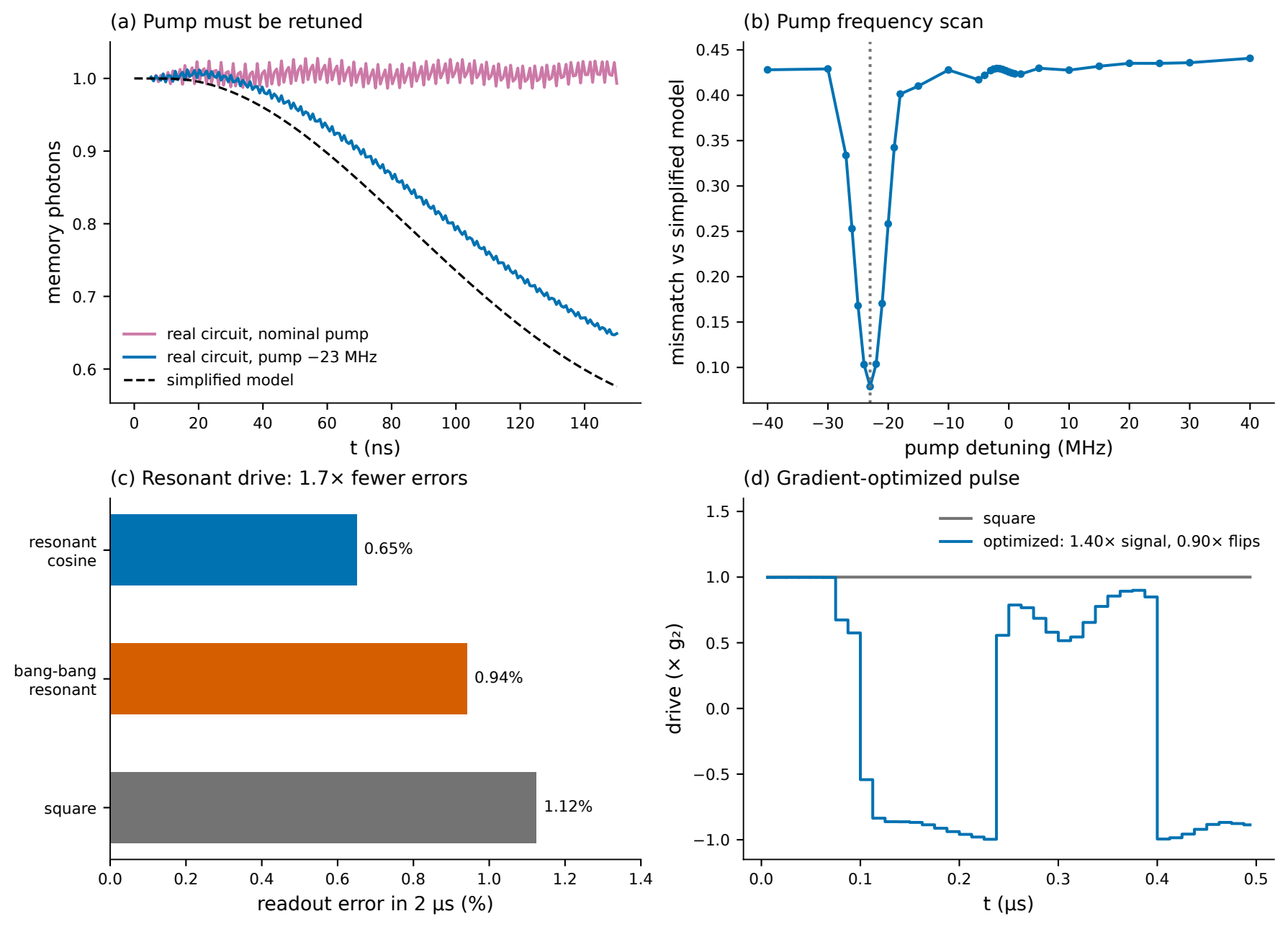
(a) Pump must be retuned
memory photons
t (ns)
real circuit, nominal pump
real circuit, pump −23 MHz
simplified model
(b) Pump frequency scan
mismatch vs simplified model
pump detuning (MHz)
(c) Resonant drive: 1.7× fewer errors
resonant cosine
bang-bang resonant
square
0.65%
0.94%
1.12%
readout error in 2 μs (%)
(d) Gradient-optimized pulse
square
optimized: 1.40× signal, 0.90× flips
drive (× g₂)
t (μs)